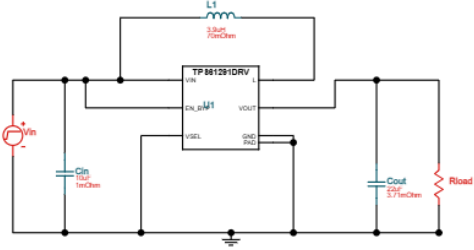


Section 1:

3.3V Boost Regulator Choice:



TPS61291
High Efficiency

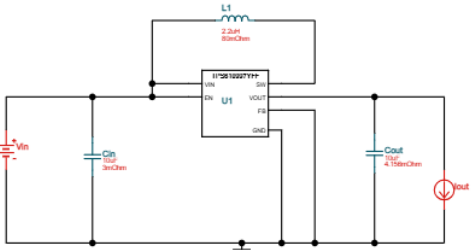
Efficiency: 91.3% BOM Cost: \$0.80 Footprint: 47 mm² BOM Count: 4 Topology: Boost
Frequency: 201.17 kHz IC Cost: \$0.62 | 1ku

[CUSTOMIZE](#) [SIMULATE](#) [EXPORT](#)

☐ Compare ⋮

I chose this regulator because it is high efficiency, only uses 4 components, it is very low cost, and has good activation frequency.

5V Boost Regulator Choice:



TPS610997
Synchronous Boost Converter with Ultra-Low Quiescent Current

Efficiency: 91.9% BOM Cost: \$0.77 Footprint: 32 mm² BOM Count: 4 Topology: Boost
Frequency: 983.14 kHz IC Cost: \$0.58 | 1ku

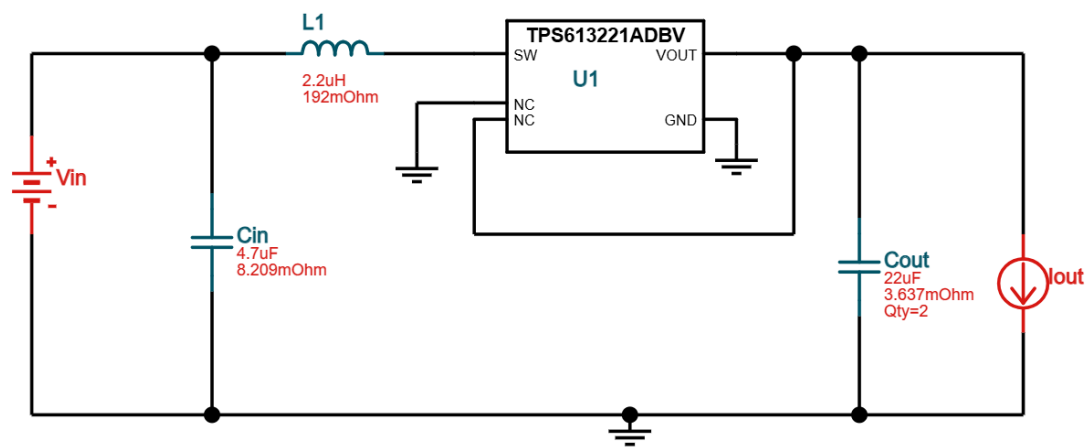
[CUSTOMIZE](#) [SIMULATE](#) [EXPORT](#)

☐ Compare ⋮

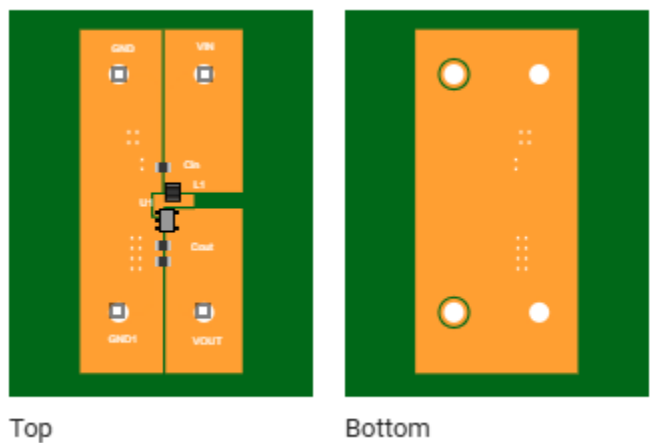
I chose this regulator because it has high efficiency, an activation frequency of less than 1 MHz which means that it will be easier to work with, and only 4 components.

Section 2

2.1 Schematic from WEBench



2.2 PCB layout from WEBench



2.3 BOM

Part	Manufacturer	Part Number	Quantity	Price (\$)	Footprint (mm ²)
U1	Texas Instruments	TPS613221ADBVR	1	0.165	14.82
Cin	Taiyo Yuden	MSAST21GAB5475MTNA01	1	0.03223	6.75
L1	MuRata	1286AS-H-2R2M	1	0.15	8.96
Cout	Taiyo Yuden	MSASL219LB5226MTNA01	2	0.07115	6.75

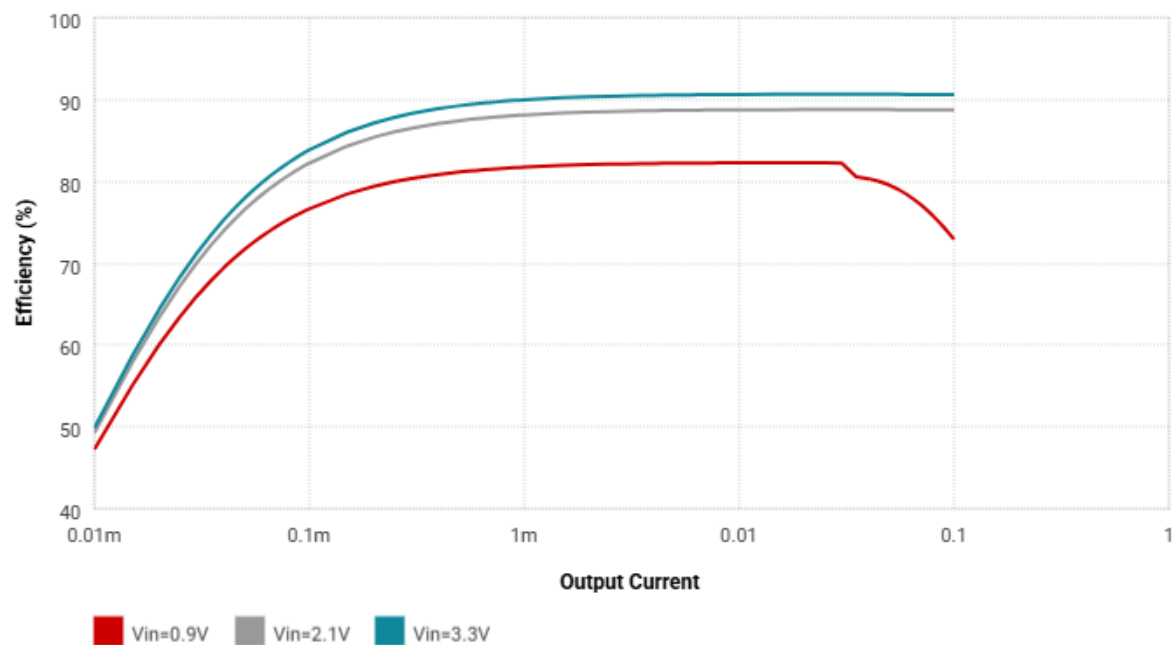
Description
Cap: 4.7 μ F Total Derated Cap: 4.7 μ F VDC: 25 V ESR: 8.21 m Ω Package: 0805
L: 2.2 μ H DCR: 192 m Ω IDC: 1.3 A
Cap: 22 μ F Total Derated Cap: 23 μ F VDC: 10 V ESR: 3.64 m Ω Package: 0805

2.4 Operating Point Table

Name	Value	Category	Description
Vout	3.3 V	System Information	Operational Output Voltage
Cin IRMS	120.63 mA	Capacitor	Input capacitor RMS ripple current
Cin Pd	119.46 μ W	Capacitor	Input capacitor power dissipation
Cout IRMS	155.36 mA	Capacitor	Output capacitor RMS ripple current
Cout Pd	43.89 μ W	Capacitor	Output capacitor power dissipation
L Ipp	466.35 mA	Inductor	Peak-to-peak inductor ripple current
Ipp percentage	246.28 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
L Pd	11.18 mW	Inductor	Inductor power dissipation
Duty Cycle	14.13%	System Information	Duty cycle
Efficiency	90.10%	System Information	Steady state efficiency
Frequency	384.69 kHz	System Information	Switching frequency
IC Tj	38.12 $^{\circ}$ C	IC	IC junction temperature
ICThetaJA	189.7 $^{\circ}$ C/W	IC	IC junction-to-ambient thermal resistance
IC Pd	42.8 mW	IC	IC power dissipation
Pout	495 mW	System Information	Total output power
Iin Avg	189.36 mA	IC	Average input current
IC Ipk	466.35 mA	IC	Peak switch current in IC
Mode	PFM	System Information	Conduction Mode

Vout p-p	10.22 mV	System Information	Peak-to-peak output ripple voltage
Vin p-p	65.12 mV	System Information	Peak-to-peak input voltage
Total Pd	54.14 mW	Power	Total Power Dissipation
FootPrint	44 mm ²	System Information	Total Foot Print Area of BOM components
Vin	2.9 V	System Information	Vin operating point
Iout	150 mA	System Information	Iout operating point
Cin Pd	119.46 μ W	Power	Input capacitor power dissipation
Cout Pd	43.89 μ W	Power	Output capacitor power dissipation
L Pd	11.18 mW	Power	Inductor power dissipation
IC Pd	42.8 mW	Power	IC power dissipation
Total BOM	\$0.49	System Information	Total BOM Cost
BOM Count	5	System Information	Total Design BOM count

2.5 Log Efficiency plot vs current



2.6 Table 2 from above

Output Voltage	Efficiency @ 150 mA	BOM Cost	Footprint	BOM Count
3.3 VDC	90.1%	0.49	44 mm ²	5

2.7 $P_{out}, P_{in}, P_{regulator}$

$$P_{out} = 495 \text{ mW}$$

$$P_{regulator} = \frac{495 \times 10^{-3} \times (1 - 0.901)}{0.901} = 54.39 \text{ mW}$$

$$P_{in} = 54.39 + 495 = 549.39 \text{ mW}$$

2.8 I_{in}

$$I_{in} = \frac{549.39 \times 10^{-3}}{2.9} \approx 189.4 \text{ mA}$$

2.9 T_j for $I_{out} = 150 \text{ mA}$

$$T_j = T_a + (R_{\theta JA} \times P_D) = 28 + (189.7 \times 54.39 \text{ m}) = 28 + 10.3178 \approx 38.318^\circ\text{C}$$

2.10

Table Values :

$$P_{out} = 495 \text{ mW}$$

$$P_{regulator} = 54.14 \text{ mW}$$

$$P_{in} = 495 \text{ mW} + 54.14 \text{ mW} = 549.14 \text{ mW}$$

$$I_{in} = 189.36 \text{ mA}$$

$$T_j = 38.12^\circ\text{C}$$

The results are closely aligned with the expected table values, with minor decimal discrepancies.

2.11 T_{j_max} from datasheet

$$T_j = 125^\circ\text{C}$$

2.12 T_j for $I_{out} = 2 \text{ A}$

$$P_{out} = 3.3 \times 2 = 6.6 \text{ W}$$

$$P_d = \frac{6.6(1 - 0.901)}{0.901} \approx 725.2 \text{ mW}$$

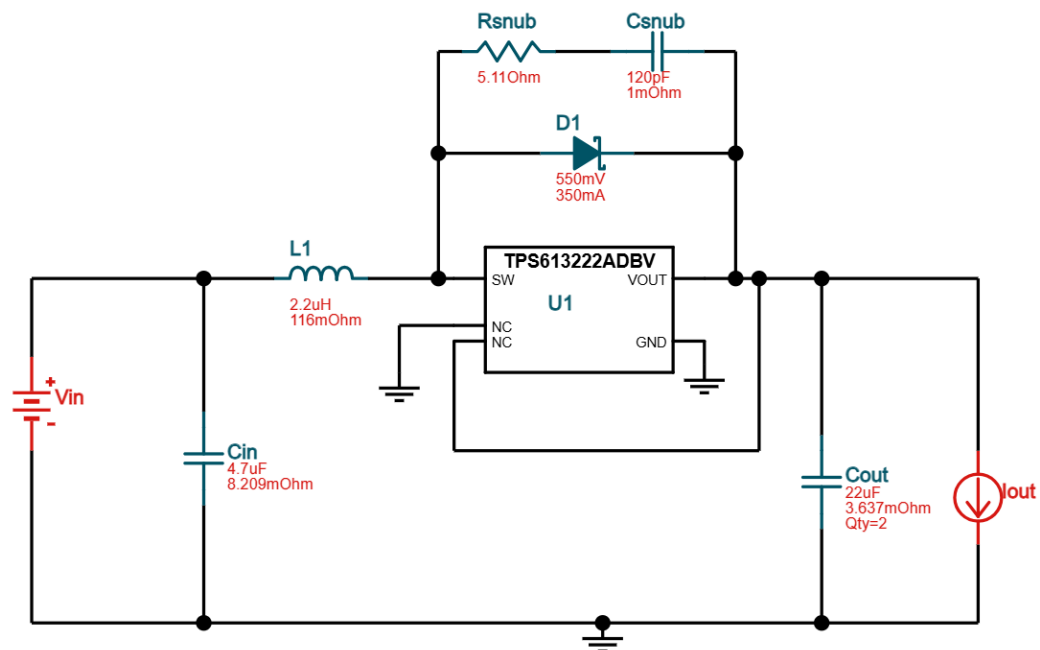
$$T_j = 28 + (189.7 \times 725.2 \text{ m}) = 28 + 137.57 \approx 165.57 \text{ Regulator is too hot.}$$

2.13: I_{LIM} (min, typ, max)

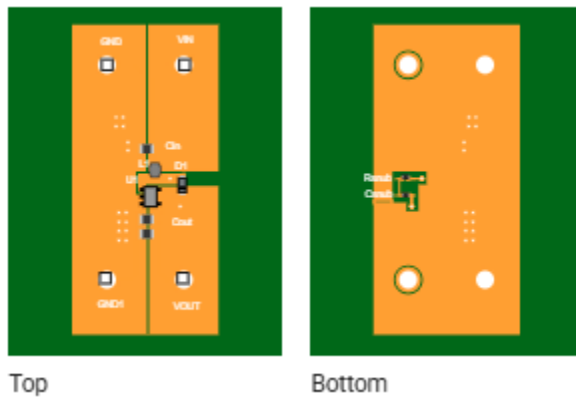
0.75, 1.20, 1.60 (A)

Section 3

3.1: Schematic from WEBench



3.2: PCB layout from WEBench



3.3: The Bill of Materials (BOM)

Part	Manufacturer	Part Number	Quantity	Price (\$)	Footprint (mm ²)
Cs _{sub}	MuRata	GRM1555C1H121JA01D	1	0.01	3
Rs _{sub}	Vishay-Dale	CRCW04025R11FKED	1	0.01	3
D1	Diodes Inc.	SD103CWS-7-F	1	0.09	8.7
U1	Texas Instruments	TPS613222ADBVR	1	0.165	14.82
C _{in}	Taiyo Yuden	MSAST21GAB5475MTNA01	1	0.03223	6.75
L1	MuRata	DFE201612E-2R2M=P2	1	0.11	7.8
C _{out}	Taiyo Yuden	MSASL219LB5226MTNA01	2	0.07115	6.75

Description

Cap: 120 pF Total Derated Cap: 120 pF VDC: 50 V ESR: 1 mΩ Package: 0402

Resistance: 5.11 Ω Tolerance: 1.0% Power: 63 mW

Type: Schottky VRRM: 20 V I_o: 350 mA

Cap: 4.7 μF Total Derated Cap: 4.7 μF VDC: 25 V ESR: 8.21 mΩ Package: 0805

L: 2.2 μH DCR: 116 mΩ IDC: 1.8 A

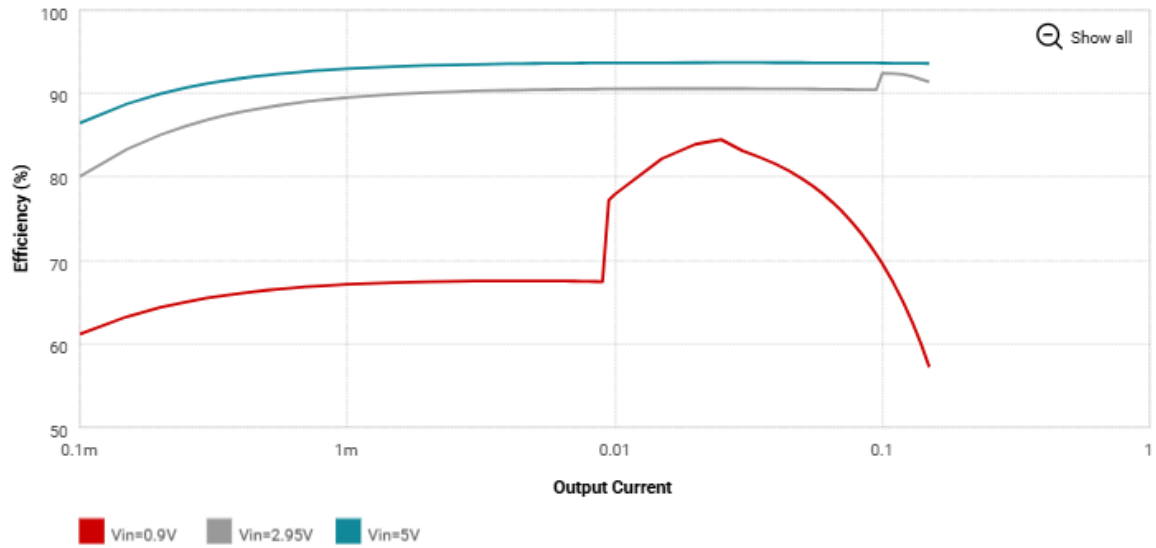
Cap: 22 μF Total Derated Cap: 15 μF VDC: 10 V ESR: 3.64 mΩ Package: 0805

3.4: Operating Point Table from WEBench

Name	Value	Category	Description
V _{out}	5 V	System Information	Operational Output Voltage
C _{in} IRMS	124.85 mA	Capacitor	Input capacitor RMS ripple current
C _{in} Pd	127.96 μW	Capacitor	Input capacitor power dissipation
C _{out} IRMS	159.49 mA	Capacitor	Output capacitor RMS ripple current
C _{out} Pd	46.26 μW	Capacitor	Output capacitor power dissipation
L I _{pp}	432.49 mA	Inductor	Peak-to-peak inductor ripple current
I _{pp} percentage	152.56 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
L Pd	9.71 mW	Inductor	Inductor power dissipation
Duty Cycle	45.60%	System Information	Duty cycle

Efficiency	91.20%	System Information	Steady state efficiency
Frequency	1.33 MHz	System Information	Switching frequency
IC Tj	41.81 °C	IC	IC junction temperature
ICThetaJA	189.7 °C/W	IC	IC junction-to-ambient thermal resistance
IC Pd	62.26 mW	IC	IC power dissipation
Pout	750 mW	System Information	Total output power
Iin Avg	283.5 mA	IC	Average input current
IC Ipk	477.22 mA	IC	Peak switch current in IC
Mode	CCM	System Information	Conduction Mode
Vout p-p	5.48 mV	System Information	Peak-to-peak output ripple voltage
Vin p-p	9.57 mV	System Information	Peak-to-peak input voltage
Total Pd	72.14 mW	Power	Total Power Dissipation
FootPrint	58 mm ²	System Information	Total Foot Print Area of BOM components
Vin	2.9 V	System Information	Vin operating point
Iout	150 mA	System Information	Iout operating point
Cin Pd	127.96 µW	Power	Input capacitor power dissipation
Cout Pd	46.26 µW	Power	Output capacitor power dissipation
L Pd	9.71 mW	Power	Inductor power dissipation
IC Pd	62.26 mW	Power	IC power dissipation
Total BOM	\$0.56	System Information	Total BOM Cost
BOM Count	8	System Information	Total Design BOM count

3.5: Log Efficiency plot versus current



3.6: Table 2 from above

Output Voltage	Efficiency @ 150 mA	BOM Cost	Footprint	BOM Count
5 VDC	91.2%	0.56	58 mm ²	8

3.7: P_{out} , P_{in} , $P_{regulator}$

$$P_{out} = 750 \text{ mW}$$

$$P_{regulator} = \frac{750 \times 10^{-3} (1 - 0.912)}{0.912} \approx 72.4 \text{ mW}$$

$$P_{in} = 750 \text{ mW} + 72.4 \text{ mW} = 822.4 \text{ mW}$$

3.8: I_{in}

$$I_{in} = \frac{822.4 \times 10^{-3}}{2.9} \approx 283.59 \text{ mA}$$

3.9 T_j for $I_{out} = 150 \text{ mA}$

$$T_j = 28 + (189.7 \times 72.3 \text{ m}) = 28 + 13.715 \approx 41.71^\circ \text{C}$$

3.10 Compare Results

Table Values:

$$P_{out} = 750 \text{ mW}$$

$$P_d = 62.26 \text{ mW}$$

$$P_{in} = 750mW + 62.26mW = 812.26 mW$$

$$I_{in} = 283.5 mA$$

$$T_j = 41.81^{\circ}C$$

The results are closely together, with some minor discrepancies.

3.11 T_j max from Datasheet:

$$T_{j_{max}} = 125^{\circ}C$$

3.12 T_j for $I_{out} = 2A$

$$P_{out} = 5 \times 2 = 10W$$

$$P_d = \frac{10 \times (1 - 0.912)}{0.912} = 964.9mW$$

$$T_j = 28 + (189.7 \times 964.9m) = 28 + 183.04 \approx 211.041^{\circ}C \text{ too hot for regulator}$$

3.13: $I_{LIM}(min, typ, max)$

1.10, 1.80, 2.50 (A)